Loan Prediction — ML Classification

Task 2 | Logistic Regression & Random Forest | Google Colab

614	13	76%	0.751
Records	Features	Accuracy	ROC-AUC (LR)

Overview: Binary classification on the Loan Prediction dataset (614 records, 13 features) to predict loan approval (Loan_Status). Pipeline: missing value imputation → Label Encoding → StandardScaler → SMOTE oversampling → model training. Two models were compared — **Logistic Regression** (Accuracy 76%, ROC-AUC 0.751) and **Random Forest** (Accuracy 76%, ROC-AUC 0.730). **Credit history** is the dominant predictor; applicant income and loan amount also rank highly. A custom threshold of 0.6 was applied to the RF model for safer approvals.

Model Comparison

Model	Precision (0)	Recall (0)	Precision (1)	Recall (1)	Accuracy	ROC-AUC
Logistic Reg.	0.75	0.49	0.77	0.91	0.76	0.751
Random Forest	0.79	0.44	0.76	0.94	0.76	0.730

Visualizations

```
plt.title("Confusion Matrix")
plt.show()
```

Fig 1 — Confusion Matrix (Random Forest)

```
plt.title("Feature Importance")
plt.show()
```

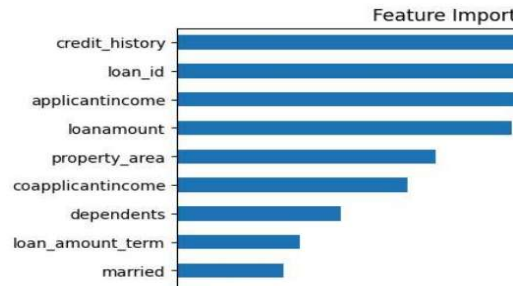

Fig 2 — Feature Importance (Random Forest)

Key Insights

- **Credit history** is the #1 predictor of loan approval
- Both models achieve **76% accuracy** on test set
- LR slightly outperforms RF on ROC-AUC (**0.751 vs 0.730**)
- RF shows higher recall for approved loans (**0.94**)
- Threshold 0.6 reduces false approvals for safer lending
- SMOTE resolved class imbalance in training data

Notebook & Tools

- **task-2_collab.ipynb** (Google Colab)
- Dataset: **loan_prediction.csv** (614 rows, 13 cols)
- pandas · sklearn · imbalanced-learn (SMOTE) · seaborn
- Steps: EDA → Clean → Encode → Scale → SMOTE → Train → Evaluate
- **Recommended: Random Forest with threshold = 0.6**